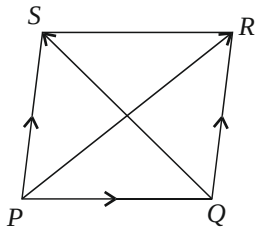
MARKING SCHEME

Senior Secondary School Examination TERM–II, 2022

MATHEMATICS (Subject Code–041)

[Paper Code: 65/1/1]

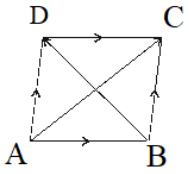
Maximum Marks: 40

Q. No.	EXPECTED ANSWER / VALUE POINTS	Marks
	SECTION-A	
1.	Find the sum of the order and the degree of the differential equation : $\left(x+\frac{dy}{dx}\right)^2=\left(\frac{dy}{dx}\right)^2+1$	2
Sol.	Given differential equation can be written as $x^2+\left(\frac{dy}{dx}\right)^2+2x\frac{dy}{dx}=\left(\frac{dy}{dx}\right)^2+1$ i.e. $x^2+2x\frac{dy}{dx}=1$; Order = 1, degree = 1 Sum of order and degree = 1 + 1 = 2	½ + ½ 1 2
2.	In a parallelogram PQRS, $\overrightarrow{PQ} = 3\hat{i} - 2\hat{j} + 2\hat{k}$ and $\overrightarrow{PS} = -\hat{i} - 2\hat{k}$. Find $\lvert\overrightarrow{PR}\rvert$ and $\lvert\overrightarrow{QS}\rvert$.	2
Sol.	$\overrightarrow{PR} = \overrightarrow{PQ} + \overrightarrow{PS} \quad \because \overrightarrow{PS} = \overrightarrow{QR}$ $= 2\hat{i} - 2\hat{j}$ $\lvert\overrightarrow{PR}\rvert = 2\sqrt{2}$ $\overrightarrow{QS} = \overrightarrow{QP} + \overrightarrow{PS} = \overrightarrow{PS} - \overrightarrow{PQ}$ $= -4\hat{i} + 2\hat{j} - 4\hat{k}$ $\lvert\overrightarrow{QS}\rvert = 6$ 	1 1 2
3.	(a) If $\frac{d}{dx}[F(x)] = \frac{\sec^4 x}{\operatorname{cosec}^4 x}$ and $F\left(\frac{\pi}{4}\right)=\frac{\pi}{4}$, then find F($x$). **OR** (b) Find : $\int \frac{\log x - 3}{(\log x)^4}\,dx$.	2
Sol.	$F(x)=\int \tan^4 x\,dx = \int \tan^2 x.(\sec^2 x - 1)\,dx$	½

	$= \int (\tan^2 x \sec^2 x - \sec^2 x + 1)\, dx$	
	$F(x) = \frac{\tan^3 x}{3} - \tan x + x + C$	1
	$x = \frac{\pi}{4},\ F(x) = \frac{\pi}{4}$ gives $C = \frac{2}{3}$	
	$F(x) = \frac{\tan^3 x}{3} - \tan x + x + \frac{2}{3}$	½
		2
	Or	
	Put $x = e^t$ or $\log x = t$ $\quad dx = e^t dt$	½
	$\therefore \int \frac{\log x - 3}{(\log x)^4} dx = \int \frac{t-3}{t^4} e^t\, dt$	
	$= \int \left(\frac{1}{t^3} - \frac{3}{t^4} \right) e^t\, dt$	½
	$= \frac{e^t}{t^3} + C$	½
	$= \frac{x}{(\log x)^3} + C$	½
		2
4.	Let A and B be two events such that P(A) = $\frac{5}{8}$, P(B) = $\frac{1}{2}$ and P(A/B) = $\frac{3}{4}$. Find the value of P(B/A).	2
Sol.	$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{3}{4} \Rightarrow P(A \cap B) = \frac{3}{8}$	1
	$P(B/A) = \frac{P(B \cap A)}{P(A)} = \frac{\frac{3}{8}}{\frac{5}{8}} = \frac{3}{5}$	1
		2
5.	Two balls are drawn at random from a bag containing 2 red balls and 3 blue balls, without replacement. Let the variable X denotes the number of red balls. Find the probability distribution of X.	2
Sol.		½

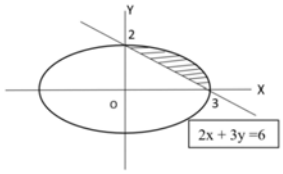
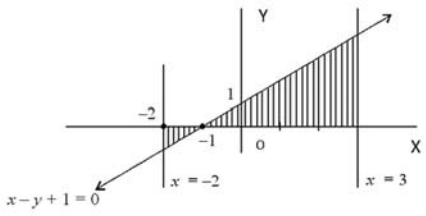
X	0	1	2
$P(X)$	$\frac{3}{5} \times \frac{2}{4} = \frac{3}{10}$	$2\left(\frac{2}{5} \times \frac{3}{4}\right) = \frac{6}{10}$	$\frac{2}{5} \times \frac{1}{4} = \frac{1}{10}$

1½

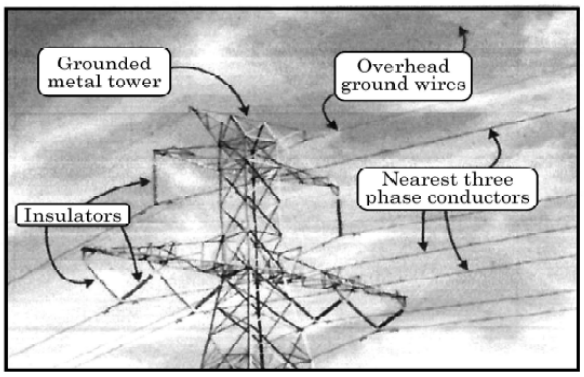
		2
6.	Find the values of λ, for which the distance of point (2, 1, λ) from plane $3x + 5y + 4z = 11$ is $2\sqrt{2}$ units.	2
Sol.	$\frac{\lvert 3\cdot 2 + 5\cdot 1 + 4\lambda - 11\rvert}{\sqrt{9+25+16}} = 2\sqrt{2}$ $\lvert 4\lambda \rvert = 20$ $\lambda = \pm 5$	1 1 2
	SECTION -B	
7.	(a) If $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$ are four non-zero vectors such that $\vec{a} \times \vec{b} = \vec{c} \times \vec{d}$ and $\vec{a} \times \vec{c} = 4\vec{b} \times \vec{d}$, then show that $(\vec{a} - 2\vec{d})$ is parallel to $(2\vec{b} - \vec{c})$ where $\vec{a} \neq 2\vec{d}$, $\vec{c} \neq 2\vec{b}$. **OR** (b) The two adjacent sides of a parallelogram are represented by $2\hat{i} - 4\hat{j} - 5\hat{k}$ and $2\hat{i} + 2\hat{j} + 3\hat{k}$. Find the unit vectors parallel to its diagonals. Using the diagonal vectors, find the area of the parallelogram also.	3
Sol. (a)	Consider $(\vec{a} - 2\vec{d}) \times (2\vec{b} - \vec{c})$ $= \vec{a} \times 2\vec{b} - \vec{a} \times \vec{c} - 4\vec{d} \times \vec{b} + 2\vec{d} \times \vec{c}$ $= 0$ $\therefore (\vec{a} - 2\vec{d}) \parallel (2\vec{b} - \vec{c})$	½ 1 1 ½ 3
(b)	**Or** Let ABCD be a parallelogram with $\overrightarrow{AB} = \overrightarrow{DC} = 2\hat{i} - 4\hat{j} - 5\hat{k}$ and $\overrightarrow{BC} = \overrightarrow{AD} = 2\hat{i} + 2\hat{j} + 3\hat{k}$ $\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC} = 4\hat{i} - 2\hat{j} - 2\hat{k}$ and $\overrightarrow{BD} = 6\hat{j} + 8\hat{k}$ $\therefore \lvert \overrightarrow{AC} \rvert = 2\sqrt{6}$ and $\lvert \overrightarrow{BD} \rvert = 10$ $\therefore$ Required unit vectors $\hat{d}_1$ and $\hat{d}_2$ are $\hat{d}_1 = \frac{2}{\sqrt{6}}\hat{i} - \frac{1}{\sqrt{6}}\hat{j} - \frac{\hat{k}}{\sqrt{6}}$ and $\hat{d}_2 = \frac{3}{5}\hat{j} + \frac{4}{5}\hat{k}$	 1 1

	Now, $\text{Area of } \|\| ABCD = \frac{1}{2}\|\vec{d}_1 \times \vec{d}_2\|$ $= \frac{1}{2}\left\|\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -2 & -2 \\ 0 & 6 & 8 \end{vmatrix}\right\|$ $= \frac{1}{2}\|-4\hat{i} - 32\hat{j} + 24\hat{k}\|$ $= \frac{1}{2}\sqrt{1616} = 2\sqrt{101}$	1 3
8.	Find the vector equation of the plane passing through the intersection of the planes $\vec{r} \cdot (2\hat{i} + 2\hat{j} - 3\hat{k}) = 7$ and $\vec{r} \cdot (2\hat{i} + 5\hat{j} + 3\hat{k}) = 9$ and through the point (2, 1, 3).	3
Sol.	Equation of the plane is $\vec{r} \cdot \left(2\hat{i} + 2\hat{j} - 3\hat{k}\right) - 7 + \lambda\left[\vec{r} \cdot \left(2\hat{i} + 5\hat{j} + 3\hat{k}\right) - 9\right] = 0$ or $\vec{r} \cdot \left[(2 + 2\lambda)\hat{i} + (2 + 5\lambda)\hat{j} + (-3 + 3\lambda)\hat{k}\right] - 7 - 9\lambda = 0$ (2, 1, 3) lies on it $\therefore\ 2(2 + 2\lambda) + 1(2 + 5\lambda) + 3(-3 + 3\lambda) - 7 - 9\lambda = 0$ $9\lambda = 10 \Rightarrow \lambda = 10/9$ Equation is $\vec{r} \cdot (38\hat{i} + 68\hat{j} + 3\hat{k}) = 153$	1 1½ ½ 3
9.	(a) Find : $\int \frac{dx}{\sqrt{x} + \sqrt[3]{x}}$. **OR** (b) Evaluate : $\int_0^{\pi/2} \frac{\cos x}{(1 + \sin x)(4 + \sin x)}\, dx$.	3
Sol. **(a)**	$I = \int \frac{dx}{\sqrt{x} + \sqrt[3]{x}}$ $= \int \frac{6y^5 dy}{y^3 + y^2}$ $\qquad x = y^6$ so that $dx = 6y^5 dy$ $= 6\int \frac{y^3}{y+1} dy$ $= 6\int \left[(y^2 - y + 1) - \frac{1}{y+1}\right] dy$	1 1

	$= 6\left[\frac{y^3}{3} - \frac{y^2}{2} + y - \log\lvert y+1\rvert\right] + C$	½
	$= 2\sqrt{x} - 3\sqrt[3]{x} + 6x^{1/6} - 6\log(x^{1/6}+1) + C$	½
		3
(b)	**Or**	
	Let $\sin x = t$, then $\cos x dx = dt$	½
	$\therefore \int_0^{\pi/2} \frac{\cos x}{(1+\sin x)(4+\sin x)} dx = \int_0^1 \frac{dt}{(1+t)(4+t)}$	½
	$= \frac{1}{3}\left[\int_0^1 \frac{1}{1+t} dt - \int_0^1 \frac{1}{4+t} dt\right]$	1
	$= \frac{1}{3}\left[\log(1+t)\Big\rvert_0^1 - \log(4+t)\Big\rvert_0^1\right]$	½
	$= \frac{1}{3}[\log 2 - \log 5 + \log 4]$	½
	or	
	$= \frac{1}{3}\log\frac{8}{5}$	
		3
10.	Find the particular solution of the differential equation $x\frac{dy}{dx} + x\cos^2\left(\frac{y}{x}\right) = y$; given that when $x = 1$, $y = \frac{\pi}{4}$.	3
Sol.	Given differential equation can be written as	
	$\frac{dy}{dx} = \frac{y}{x} - \cos^2\frac{y}{x}$ ... (1)	½
	Let $y = vx$ $\therefore \frac{dy}{dx} = v + x\frac{dv}{dx}$	½
	(1) Becomes $v + x\frac{dv}{dx} = v - \cos^2 v$	
	$\sec^2 v\, dv = -\frac{dx}{x}$	
	Integrating both sides we get	
	$\tan v = -\log\lvert x\rvert + c$	½
	$\tan\frac{y}{x} = -\log\lvert x\rvert + c$	
	$x = 1,\ y = \frac{\pi}{4} \Rightarrow c = 1$	1

	∴ Particular solution is $\tan\frac{y}{x} = -\log\lvert x\rvert + 1$	½
		3
	SECTION-C	
11.	(a) Using integration, find the area of the region $\{(x, y) : 4x^2 + 9y^2 \le 36, 2x + 3y \ge 6\}$. **OR** (b) Using integration, find the area of the region bounded by lines $x - y + 1 = 0$, $x = -2$, $x = 3$ and x-axis.	4
Sol.		
	Correct figure	1
	Clearly points of intersection are (3, 0) and (0, 2)	½
	Required area $= \frac{2}{3}\int_0^3 \sqrt{9-x^2}\,dx - \frac{2}{3}\int_0^3 (3-x)\,dx$	1
	$= \frac{2}{3}\left[\frac{x}{2}\sqrt{9-x^2} + \frac{9}{2}\sin^{-1}\frac{x}{3} + \frac{(3-x)^2}{2}\right]_0^3$	1
	$= \frac{2}{3}\left[\frac{9\pi}{4} - \frac{9}{2}\right]$ *or* $\frac{3\pi}{2} - 3$	½
		4
	Or	
		
	Correct figure	1
	Required area $= -\int_{-2}^{-1}(x+1)\,dx + \int_{-1}^{3}(x+1)\,dx$	1

	$= -\frac{(x+1)^2}{2}\Big\|_{-2}^{-1} + \frac{(x+1)^2}{2}\Big\|_{-1}^{3}$ $= \frac{1}{2} + 8$ $= \frac{17}{2}$	1 1 4
12.	A card from a pack of 52 playing cards is lost. From the remaining cards, 2 cards are drawn at random without replacement, and are found to be both aces. Find the probability that lost card being an ace.	4
Sol.	E_1 : Lost card is an ace E_2 : Lost card is not an ace A: 2 ace cards are drawn $P(E_1) = \frac{1}{13}$ $\quad P(E_2) = \frac{12}{13}$ $P(A/E_1) = \frac{{}^3C_2}{{}^{51}C_2}$ $\quad P(A/E_2) = \frac{{}^4C_2}{{}^{51}C_2}$ $P(E_1/A) = \frac{P(E_1)P(A/E_1)}{P(E_1)P(A/E_1) + P(E_2)P(A/E_2)}$ $= \frac{\frac{1}{13}\frac{{}^3C_2}{{}^{51}C_2}}{\frac{1}{13}\cdot\frac{{}^3C_2}{{}^{51}C_2} + \frac{12}{13}\cdot\frac{{}^4C_2}{{}^{51}C_2}}$ $= \frac{3}{75}$ or $\frac{1}{25}$	½ 1 1 1 + ½ 4
13.	Evaluate : $\int_0^{\pi} \frac{x}{1+\sin x}\,dx$.	4
Sol.	$I = \int_0^{\pi} \frac{x}{1+\sin x}dx = \int_0^{\pi} \frac{\pi - x}{1+\sin(\pi - x)}dx = \pi\int_0^{\pi} \frac{1}{1+\sin x}dx - \int_0^{\pi} \frac{x}{1+\sin x}dx$ $\therefore\ 2I = \pi\int_0^{\pi} \frac{dx}{1+\sin x}$ $\therefore\ I = \frac{\pi}{2}\int_0^{\pi} \frac{dx}{1+\sin x} = \frac{\pi}{2}\int_0^{\pi} \frac{1}{1+\cos\left(\frac{\pi}{2} - x\right)}dx$	1 ½ ½

	$= \frac{\pi}{2}\int_0^{\pi} \frac{1}{2\cos^2\left(\frac{\pi}{4}-\frac{x}{2}\right)}dx$	
	$= \frac{\pi}{4}\left[-2\tan\left(\frac{\pi}{4}-\frac{x}{2}\right)\right]_0^{\pi}$	1
	$= \frac{\pi}{4}[2-(-2)] = \pi$	1
		4
14.	Electrical transmission wires which are laid down in winters are stretched tightly to accommodate expansion in summers.  Two such wires lie along the following lines : $l_1 : \frac{x+1}{3} = \frac{y-3}{-2} = \frac{z+2}{-1}$ $l_2 : \frac{x}{-1} = \frac{y-7}{3} = \frac{z+7}{-2}$ Based on the given information, answer the following questions :	
	(i) Are the lines l_1 and l_2 coplanar ? Justify your answer.	2
	(ii) Find the point of intersection of the lines l_1 and l_2.	2
Sol.	*(i)* Consider $\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \end{vmatrix}$	
	$= \begin{vmatrix} +1 & 4 & -5 \\ 3 & -2 & -1 \\ -1 & 3 & -2 \end{vmatrix}$	1
	$= 1(+7) - 4(-7) - 5(7)$	
	$= 0$	
	$\therefore$ lines l_1 and l_2 are coplanar	1
	(ii) Any point on $l_1 : (3\lambda - 1, -2\lambda + 3, -\lambda - 2)$	1
	Substituting in equation of l_2,	

	$\frac{3\lambda-1}{-1}=\frac{-2\lambda+3-7}{3}$ $\Rightarrow 9\lambda-3=2\lambda+4$ $\Rightarrow 7\lambda=7\Rightarrow\lambda=1$ Point is $(2,1,-3)$	½ ½
		2 + 2

MARKING SCHEME

Senior Secondary School Examination TERM–II, 2022

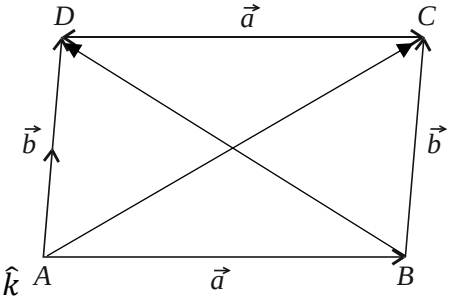
MATHEMATICS (Subject Code–041)

[Paper Code: 65/2/1]

Maximum Marks: 40

Q. No.	EXPECTED ANSWER / VALUE POINTS	Marks
	SECTION – A	
1.	Find the product of the order and the degree of the differential equation $\left[\frac{d}{dx}(xy^2)\right] \cdot \frac{dy}{dx} + y = 0$.	2
Sol.	Given differential equation can be written as $2xy\left(\frac{dy}{dx}\right)^2 + y^2\frac{dy}{dx} + y = 0$ Order = 1, Degree = 2	1
	Order × degree = 1 × 2 = 2	1
		2
2. (a)	Find : $\int \frac{\sin 3x}{\sin x}\,dx$	2
Sol.	$\int \frac{\sin 3x}{\sin x}dx = \int \frac{3\sin x - 4\sin^3 x}{\sin x}dx$	½
	$= \int [3 - 4\frac{(1-\cos 2x)}{2}]dx$	½
	$= \int (1 + 2\cos 2x)dx$	
	$= x + \sin 2x + C$	1
		2
	Or	
2. (b)	Evaluate : $\int_0^{\frac{1}{2}\log 3} \frac{e^x}{e^{2x}+1}\,dx$	2
Sol.	Let $e^x = t,\ e^x dx = dt$	½

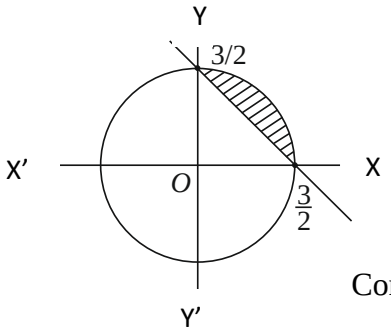
	$\therefore \int_0^{\frac{1}{2}\log 3} \frac{e^x}{e^{2x}+1}dx = \int_1^{\sqrt{3}} \frac{dt}{t^2+1}$ ½ $= \tan^{-1} t\Big\|_1^{\sqrt{3}}$ ½ $= \frac{\pi}{3} - \frac{\pi}{4} = \frac{\pi}{12}$ ½	2
3.	$\vec{a}$ and $\vec{b}$ are two unit vectors such that $\|2\vec{a} + 3\vec{b}\| = \|3\vec{a} - 2\vec{b}\|$. Find the angle between $\vec{a}$ and $\vec{b}$.	2
Sol.	$\|2\vec{a} + 3\vec{b}\| = \|3\vec{a} - 2\vec{b}\|$ $\Rightarrow \|2\vec{a} + 3\vec{b}\|^2 = \|3\vec{a} - 2\vec{b}\|^2$ $\Rightarrow 4\|\vec{a}\|^2 + 12\,\vec{a}.\vec{b} + 9\|\vec{b}\|^2 = 9\|\vec{a}\|^2 - 12\,\vec{a}.\vec{b} + 4\|\vec{b}\|^2$ 1 As $\|\vec{a}\| = \|\vec{b}\| = 1$ $\therefore 24\,\vec{a}.\vec{b} = 5\|\vec{a}\|^2 - 5\|\vec{b}\|^2 = 0 \Rightarrow \vec{a}.\vec{b} = 0$ ½ So, $\vec{a} \perp \vec{b}$ or Angle between them is $\frac{\pi}{2}$ ½	2
4.	A pair of dice is thrown. It is given that the sum of numbers appearing on both dice is an even number. Find the probability that the number appearing on at least one die is 3.	2
Sol.	A: number appearing on at least one die is 3 B: sum of numbers appearing on both dice is even Clearly, $A \cap B = \{(3,1),(3,5),(1,3),(5,3),(3,3)\}$ ½ $P\left(\frac{A}{B}\right) = \frac{P(A\cap B)}{P(B)} = \frac{5/36}{18/36}$ 1 $= \frac{5}{18}$ ½	2
5.	Probabilities of A and B solving a specific problem are $\frac{2}{3}$ and $\frac{3}{5}$, respectively. If both of them try independently to solve the problem, then find the probability that the problem is solved.	2
Sol.	P (Problem is solved) $= 1 - P$ (Problem not solved) $= 1 - P(\bar{A})P(\bar{B})$	1

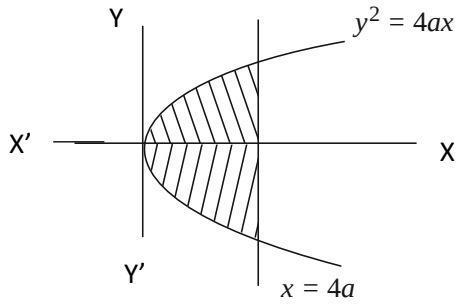
	$= 1 - \frac{1}{3} \cdot \frac{2}{5}$ $= \frac{13}{15}$	1 2
6.	Write the cartesian equation of the line PQ passing through point P(2, 2, 1) and Q(5, 1, – 2). Hence, find the y-coordinate of the point on the line PQ whose z-coordinate is – 2.	2
Sol.	Required equation of line is given by $\frac{x-2}{3} = \frac{y-2}{-1} = \frac{z-1}{-3}$ putting $z = -2$, we get $\frac{y-2}{-1} = \frac{-3}{-3} = 1$ $y - 2 = -1 \Rightarrow y = 1$	1 ½ ½ 2
	SECTION – B	
7.	ABCD is a parallelogram such that $\overrightarrow{AC} = \hat{i} + \hat{j}$ and $\overrightarrow{BD} = 2\hat{i} + \hat{j} + \hat{k}$. Find $\overrightarrow{AB}$ and $\overrightarrow{AD}$. Also, find the area of the parallelogram ABCD.	3
Sol.	Let $\overrightarrow{AB} = \vec{a}$ and $\overrightarrow{AD} = \vec{b}$ $\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC} = \vec{a} + \vec{b} = \hat{\imath} + \hat{\jmath}$ $\overrightarrow{BD} = \overrightarrow{BC} + \overrightarrow{CD} = \vec{b} - \vec{a} = 2\hat{\imath} + \hat{\jmath} + \hat{k}$  Adding we get, $2\overrightarrow{AD} = \overrightarrow{AC} + \overrightarrow{BD} = 3\hat{\imath} + 2\hat{\jmath} + \hat{k}$ $\Rightarrow \overrightarrow{AD} = \frac{3}{2}\hat{\imath} + \hat{\jmath} + \frac{1}{2}\hat{k}$ Subtracting, we get $2\overrightarrow{AB} = \overrightarrow{AC} - \overrightarrow{BD} = -\hat{\imath} - \hat{k} \Rightarrow \overrightarrow{AB} = -\frac{1}{2}\hat{\imath} - \frac{1}{2}\hat{k}$ $\left\lvert \overrightarrow{AC} \times \overrightarrow{BD} \right\rvert = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 1 & 1 & 0 \\ 2 & 1 & 1 \end{vmatrix} = \hat{\imath} - \hat{\jmath} - \hat{k}$ Area $= \frac{1}{2}\left\lvert \overrightarrow{AC} \times \overrightarrow{BD} \right\rvert$ $= \frac{\sqrt{3}}{2}$	1 1 1

		3
8.(a)	Evaluate : $\int_0^1 \tan^{-1} x \, dx$	3
Sol.	Consider $\int (\tan^{-1} x)\, dx = \tan^{-1} x \cdot x - \int \frac{1}{1+x^2} \cdot x \, dx$ $= x \tan^{-1} x - \frac{1}{2}\log(1+x^2)$ $\int_0^1 (\tan^{-1} x)dx = x \tan^{-1} x - \frac{1}{2} log(1+x^2)\Big]_0^1$ $= \frac{\pi}{4} - \frac{1}{2}\log 2$	1 1 1 3
	Or	
8.(b)	Find : $\int \frac{2x}{x^2+3x+2} dx$	3
Sol.	$I = \int \frac{2x \cdot dx}{x^2+3x+2} = \int \frac{2x}{(x+1)(x+2)} dx$ $= \int \left(\frac{-2}{x+1} + \frac{4}{x+2} \right) dx$ using partial fraction $= -2\log\lvert x+1\rvert + 4\log\lvert x+2\rvert + C$	½ 1½ 1 3
9.	Find the particular solution of the differential equation $(y + 3x^2)\frac{dx}{dy} = x$, given that y = 1, when x = 1.	3
Sol.	Given differential equation can be written as $x\frac{dy}{dx} - y = 3x^2$ or $\frac{dy}{dx} - \frac{1}{x} y = 3x$ I.F = $e^{\int -\frac{1}{x} dx} = e^{-\log x} = -x^{-1} = \frac{1}{x}$ Solution is $y \cdot \frac{1}{x} = \int 3x \frac{1}{x} dx + C$ $\frac{y}{x} = 3x + C$	½ 1 ½ ½

	$x=1,\ y=1$ gives $C=-2$ Particular solution is $\frac{y}{x}=3x-2$ or $y=3x^2-2x$	½ 3
10.(a)	Find the equation of the plane passing through points (2, 1, 0), (3, – 2, – 2) and (1, 1, – 7). Also, obtain its distance from the origin.	3
Sol.	Equation of plane is given by $\begin{vmatrix} x-2 & y-1 & z \\ 1 & -3 & -2 \\ -1 & 0 & -7 \end{vmatrix}=0$ $21(x-2)+9(y-1)-3z=0$ i.e., $7x+3y-z=17$ Distance of plane from origin is $d=\frac{\lvert 0+0-0-17\rvert}{\sqrt{59}}=\frac{17}{\sqrt{59}}$	1½ 1 ½ 3
	Or	
10.(b)	Find the distance between the lines $x=\frac{y-1}{2}=\frac{z-2}{3}$ and $x+1=\frac{y+2}{2}=\frac{z-1}{3}$.	3
Sol.	For lines $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$ and $\frac{x+1}{1}=\frac{y+2}{2}=\frac{z-1}{3}$ Let $\overrightarrow{a_1}=\hat{j}+2\hat{k}, \vec{b}=\hat{\imath}+2\hat{j}+3\hat{k}$ $\overrightarrow{a_2}=-\hat{\imath}-2\hat{j}+\hat{k}, \vec{b}=\hat{\imath}+2\hat{j}+3\hat{k}$ Clearly lines are parallel Hence, Shortest distance or distance is given by $\frac{\lvert(\overrightarrow{a_2}-\overrightarrow{a_1})\times\vec{b}\rvert}{\lvert\vec{b}\rvert}$ $\overrightarrow{a_2}-\overrightarrow{a_1}=-\hat{\imath}-3\hat{j}-\hat{k}$ $(\overrightarrow{a_2}-\overrightarrow{a_1})\times\vec{b}=\begin{vmatrix} \hat{\imath} & \hat{j} & \hat{k} \\ -1 & -3 & -1 \\ 1 & 2 & 3 \end{vmatrix}$	½ ½

	$= -7\hat{i} + 2\hat{j} + \hat{k}$ Required distance = $\frac{\sqrt{49+4+1}}{\sqrt{1+4+9}} = \frac{\sqrt{27}}{\sqrt{7}}$ or $\frac{3\sqrt{21}}{7}$	1 1 3
	SECTION – C	
11.	Find the distance of the point $(-1, -5, -10)$ from the point of intersection of the line $\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12}$ and the plane $x - y + z = 5$.	4
Sol.	General point of the line is $(3\lambda + 2,\ 4\lambda - 1,\ 12\lambda + 2)$ This will be the point of intersection with the plane, if $(3\lambda + 2) - (4\lambda - 1) + (12\lambda + 2) = 5$ $\Rightarrow 11\lambda + 5 = 5,\ \Rightarrow \lambda = 0$ Thus, the point of intersection is (2, –1, 2) Distance = $\sqrt{(2+1)^2 + (-1+5)^2 + (2+10)^2}$ $= \sqrt{9+16+144}$ $= \sqrt{169} = 13$	1 1 1 1 4
12.	Evaluate : $\int_0^1 x\,(1-x)^n\,dx$	4

Sol.	$I = \int_0^1 x(1-x)^n dx$ $= \int_0^1 (1-x)[1-(1-x)]^n dx$ [using property] $= \int_0^1 x^n (1-x)\, dx$ $= \int_0^1 x^n dx - \int_0^1 x^{n+1} dx$ $= \left[\frac{x^{n+1}}{n+1}\right]_0^1 - \left[\frac{x^{n+2}}{n+2}\right]_0^1$ $= \frac{1}{n+1} - \frac{1}{n+2}$ Or $\frac{1}{(n+1)(n+2)}$	1 1 1 1 4
13.(a)	Using integration, find the area of the smaller region enclosed by the curve $4x^2 + 4y^2 = 9$ and the line $2x + 2y = 3$.	4
Sol.	Clearly point of intersection are $\left(\frac{3}{2}, 0\right)$ & $\left(0, \frac{3}{2}\right)$  Correct Figure Required area = $\int_0^{3/2} \sqrt{\frac{9}{4} - x^2}\, dx - \int_0^{3/2} \left(\frac{3}{2} - x\right) dx$ $= \frac{x}{2}\sqrt{\frac{9}{4} - x^2} + \frac{9}{8}\sin^{-1}\frac{2x}{3}\Big]_0^{3/2} + \frac{\left(\frac{3}{2} - x\right)^2}{2}\Big]_0^{3/2}$ $= \frac{9\pi}{16} - \frac{9}{8}$	½ 1 1 1 ½ 4
	Or	

13.(b)	If the area of the region bounded by the curve $y^2 = 4ax$ and the line $x = 4a$ is $\frac{256}{3}$ sq. units, then using integration, find the value of a, where $a > 0$.	4
Sol.	Given area = $\frac{256}{3}$  Correct Figure Area of Shaded region = $2\int_0^{4a} \sqrt{4ax}\, dx$ $= 8\sqrt{a}\, \frac{x^{3/2}}{3}\Big\|_0^{4a}$ $= \frac{64a^2}{3}$ $\frac{64a^2}{3} = \frac{256}{3}$ $\Rightarrow a^2 = 4$ gives $a = 2$ (as $a > 0$)	1 1 1 ½ ½ 4

14.

Case-Study Based Question

At the start of a cricket match, a coin is tossed and the team winning the toss has the opportunity to choose to bat or bowl. Such a coin is unbiased with equal probabilities of getting head and tail.



Based on the above information, answer the following questions :

(a) If such a coin is tossed 2 times, then find the probability distribution of number of tails. — 2

(b) Find the probability of getting at least one head in three tosses of such a coin. — 2

Sol.

(a) Let X denote the number of tails: — ½

X	0	1	2
$P(X)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

— 1½

(b) S = {HHH, HHT, HTH, HTT, THH, THT, TTH, TTT}

P (at least one head) = 1–P (no head) — 1

$$= 1-\frac{1}{8}$$

$$=\frac{7}{8}$$

— 1

2+2

* * *

MARKING SCHEME

Senior Secondary School Examination TERM–II, 2022

MATHEMATICS (Subject Code — 041)

[Paper Code — 65/3/1]

Maximum Marks: 40

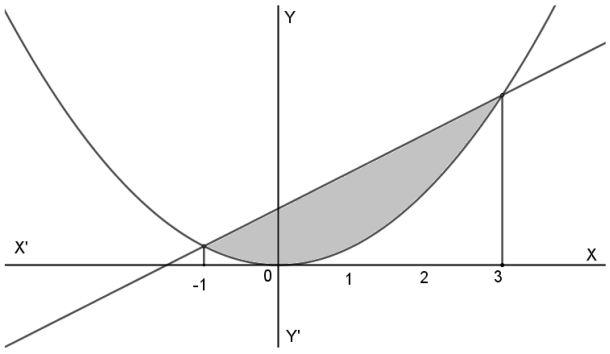
Q. No.	EXPECTED ANSWER / VALUE POINTS	Marks
	SECTION—A	
1.	Find : $\int \frac{dx}{x^2-6x+13}$	2
Sol.	$\int \frac{dx}{x^2-6x+13}$ $= \int \frac{dx}{(x-3)^2+(2)^2}$ $= \frac{1}{2}tan^{-1}\frac{(x-3)}{2} + C$	1 1 2
2.	Find the general solution of the differential equation : $e^{dy/dx} = x^2$.	2
Sol.	$e^{dy/dx} = x^2$ $\frac{dy}{dx} = \log x^2 = 2\log x$ $\int dy = 2\int \log x\, dx$ $y = 2[\log x \times x - \int 1 dx]$ $y = 2[x\, log\, x - x] + C$	½ ½ ½ ½ 2
3.	Write the projection of the vector ($\vec{b} + \vec{c}$) on the vector $\vec{a}$, where $\vec{a} = 2\hat{i} - 2\hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - 2\hat{k}$ and $\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$.	2
Sol.	$\vec{b} + \vec{c} = 3\hat{\imath} + \hat{\jmath} + 2\hat{k}$ Projection of $\vec{b} + \vec{c}$ on $\vec{a} = \frac{(\vec{b}+\vec{c}).\vec{a}}{\lvert\vec{a}\rvert}$ $= \frac{(3\hat{i} + \hat{j} + 2\hat{k}) \cdot (2\hat{i} - 2\hat{j} + \hat{k})}{\sqrt{9}}$ $= \frac{6-2+2}{3} = \frac{6}{3} = 2$	½ 1 ½ 2

4.	If the distance of the point (1, 1, 1) from the plane $x - y + z + \lambda = 0$ is $\frac{5}{\sqrt{3}}$, find the value(s) of λ.	2
Sol.	Distance of point (1, 1, 1) from $x - y + z + \lambda = 0$ is $\frac{\lvert 1+\lambda \rvert}{\sqrt{3}}$ $\frac{\lvert 1+\lambda \rvert}{\sqrt{3}} = \frac{5}{\sqrt{3}} \Rightarrow 1 + \lambda = \pm 5$ $\lambda = -6$ or $+4$	½ 1 ½ 2
5.	Two cards are drawn successively with replacement from a well shuffled pack of 52 cards. Find the probability distribution of the number of spade cards.	2
Sol.	Let X denotes the number of spades $p = \frac{13}{52} = \frac{1}{4}, q = \frac{3}{4}$ X: 0 | 1 | 2 P(X): $\frac{3}{4}.\frac{3}{4} = \frac{9}{16}$ | $2.\frac{3}{4}.\frac{1}{4} = \frac{6}{16}$ | $\frac{1}{4}.\frac{1}{4} = \frac{1}{16}$	½ 1½ 2
6.(a)	A pair of dice is thrown and the sum of the numbers appearing on the dice is observed to be 7. Find the probability that the number 5 has appeared on atleast one die.	2
Sol.	A: sum is 7 B: 5 has appeared at least on one die $A \cap B = \{(2,5), (5,2)\}$ $P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} = \frac{2/36}{6/36} = \frac{1}{3}$	½ ½ 1 2
	OR	
6.(b)	The probability that A hits the target is $\frac{1}{3}$ and the probability that B hits it, is $\frac{2}{5}$. If both try to hit the target independently, find the probability that the target is hit.	2

Sol.	$E_1 = A$ hits the target $E_2 = B$ hits the target P(target is hit) = 1 – P(target is not hit) $= 1 - \left(\frac{2}{3}.\frac{3}{5}\right) = \frac{3}{5}$	1 1 2
	SECTION—B	
7.	Evaluate : $\int_0^{2\pi} \frac{dx}{1+e^{\sin x}}$	3
Sol.	$I = \int_0^{2\pi} \frac{dx}{1+e^{\sin x}}$ (1) $I = \int_0^{2\pi} \frac{dx}{1+e^{\sin(2\pi - x)}}$ $I = \int_0^{2\pi} \frac{dx}{1+e^{-\sin x}}$ $I = \int_0^{2\pi} \frac{e^{\sin x}}{1+e^{\sin x}} dx$ (2) Adding (1) and (2) $2I = \int_0^{2\pi} 1 dx \Rightarrow 2I = 2\pi$ $I = \pi$	½ ½ ½ 1 ½ 3
8.(a)	Find the particular solution of the differential equation $x\frac{dy}{dx} - y = x^2 \cdot e^x$, given y(1) = 0.	3
Sol.	$\frac{dy}{dx} - \frac{y}{x} = \left(\frac{x^2}{x}\right)e^x = xe^x$ $I \cdot F = e^{\int -\frac{1}{x}dx} = e^{-\log x} = e^{\log x^{-1}} = \frac{1}{x}$ Solution is $y \times \frac{1}{x} = \int e^x dx$ $\frac{y}{x} = e^x + C$ $y = xe^x + Cx$ $C = -e$	½ 1 ½ ½

	Particular solution is $y = xe^x - ex$	½ 3
	OR	
8.(b)	Find the general solution of the differential equation $x\frac{dy}{dx} = y(\log y - \log x + 1)$.	3
Sol.	$x\frac{dy}{dx} = y[\log y - \log x + 1]$ $\frac{dy}{dx} = \frac{y}{x}\left[\log\frac{y}{x} + 1\right]$ $y = vx$ $\frac{dy}{dx} = v + x\frac{dv}{dx}$ $v + x\frac{dv}{dx} = v[\log v + 1]$ $x\frac{dv}{dx} = v\log v$ $\int\frac{dv}{v\log v} = \int\frac{1}{x}dx$ $\log\lvert\log v\rvert = \log\lvert x\rvert + \log C$ $\log\left(\frac{y}{x}\right) = Cx$	½ ½ ½ ½ ½ ½ 3
9.(a)	The two adjacent sides of a parallelogram are represented by vectors $2\hat{i} - 4\hat{j} + 5\hat{k}$ and $\hat{i} - 2\hat{j} - 3\hat{k}$. Find the unit vector parallel to one of its diagonals. Also, find the area of the parallelogram.	3
Sol.	One diagonal of the parallelogram $= (2\hat{i} - 4\hat{j} + 5\hat{k}) + (\hat{i} - 2\hat{j} - 3\hat{k})$ $= 3\hat{i} - 6\hat{j} + 2\hat{k}$ Unit vector parallel to the diagonal $= \frac{3\hat{i} - 6\hat{j} + 2\hat{k}}{\sqrt{9+36+4}}$ $= \frac{3}{7}\hat{i} - \frac{6}{7}\hat{j} + \frac{2}{7}\hat{k}$	1 1

	Vector area of parallelogram $= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -4 & 5 \\ 1 & -2 & -3 \end{vmatrix}$ $= \hat{i}(22) - \hat{j}(-11) + \hat{k}(0)$ $= 22\hat{i} + 11\hat{j}$ Area $= \sqrt{484+121} = \sqrt{605} = 11\sqrt{5}$	1 3
	OR	
9.(b)	If $\vec{a} = 2\hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = -\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{c} = 3\hat{i} + \hat{j}$ are such that the vector $(\vec{a} + \lambda\vec{b})$ is perpendicular to vector $\vec{c}$, then find the value of λ.	3
Sol.	$(\vec{a} + \lambda\vec{b}).\vec{c} = 0 \Rightarrow [(2-\lambda)\hat{i} + (2+2\lambda)\hat{j} + (3+\lambda)\hat{k}].(3\hat{i} + \hat{j}) = 0$ $\Rightarrow 3(2-\lambda) + (2+2\lambda)\cdot 1 = 0$ $\Rightarrow -3\lambda + 2\lambda + 6 + 2 = 0$ $\Rightarrow \lambda = 8$	½+1 1 ½ 3
10.	Show that the lines : $\frac{1-x}{2} = \frac{y-3}{4} = \frac{z}{-1}$ and $\frac{x-4}{3} = \frac{2y-2}{-4} = z-1$ are coplanar.	3
Sol.	Equations of given lines in standard form are $\frac{x-1}{-2} = \frac{y-3}{4} = \frac{z}{-1}$ and $\frac{x-4}{3} = \frac{y-1}{-2} = \frac{z-1}{1}$ Lines are coplanar if $\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$ LHS $= \begin{vmatrix} 3 & -2 & 1 \\ -2 & 4 & -1 \\ 3 & -2 & 1 \end{vmatrix}$ $= 3(4-2) + 2(-2+3) + 1(4-12)$ $= 6 + 2 - 8$ $= 0$	½ ½ ½ ½

	$\Rightarrow$ Coplanar.	1
		3
	SECTION—C	
11.	Find the area of the region bounded by curve $4x^2$ = y and the line y = $8x + 12$, using integration.	4
Sol.	 Correct Figure	1
	Point of intersection $4x^2 = 8x + 12$ $x^2 - 2x - 3 = 0$ $(x-3)(x+1) = 0$ $x = 3, -1$	½
	Area $= \int_{-1}^{3}[(8x + 12) - 4x^2]dx$	1
	$= 4x^2 + 12x - \frac{4}{3}x^3\Big]_{-1}^{3}$	1
	$= 36 + 36 - 36 - \left(4 - 12 + \frac{4}{3}\right)$ $= 44 - \frac{4}{3} = \frac{128}{3}$	½
		4
12.(a)	Find : $\int \frac{x^2}{(x^2+1)(3x^2+4)}\,dx$	4
Sol.	$\int \frac{x^2 dx}{(x^2+1)(3x^2+4)}$ Let $x^2 = t$	½
	$\frac{t}{(t+1)(3t+4)} = \frac{-1}{t+1} + \frac{4}{3t+4}$ (by Partial fraction)	1

	$\int \frac{x^2}{(x^2+1)(3x^2+4)}dx = \int -\frac{1}{x^2+1}dx + \int \frac{4}{3x^2+4}dx$	½
	$= -tan^{-1}x + \frac{4}{3} \times \frac{\sqrt{3}}{2} tan^{-1}\frac{\sqrt{3}x}{2} + C$	1½
	$= -tan^{-1}x + \frac{2}{\sqrt{3}} tan^{-1}\frac{\sqrt{3}x}{2} + C$	½
		4
	OR	
12.(b)	Evaluate : $\int_{-2}^{1} \sqrt{5-4x-x^2}\, dx$	4
Sol.	$\int_{-2}^{1} \sqrt{5-4x-x^2}dx$	
	$= \int_{-2}^{1} \sqrt{9-(x+2)^2}dx$	1½
	$= \frac{x+2}{2}\sqrt{9-(x+2)^2} + \frac{9}{2}\sin^{-1}\frac{x+2}{3}\Big]_{-2}^{1}$	1½
	$= 0 + \frac{9}{2}.\frac{\pi}{2} - 0$	
	$= \frac{9\pi}{4}$	1
		4
13.	Find the distance of the point (1, –2, 9) from the point of intersection of the line $\vec{r} = 4\hat{i} + 2\hat{j} + 7\hat{k} + \lambda(3\hat{i} + 4\hat{j} + 2\hat{k})$ and the plane $\vec{r} \cdot (\hat{i} - \hat{j} + \hat{k}) = 10$.	4
Sol.	Given line is $\vec{r} = (4\hat{\imath} + 2\hat{\jmath} + 7\hat{k}) + \lambda(3\hat{\imath} + 4\hat{\jmath} + 2\hat{k})$	
	Any point on the line is	
	$(4+3\lambda,\ 2+4\lambda,\ 7+2\lambda)$	1
	This point lies on the plane $\vec{r}.(\hat{\imath} - \hat{\jmath} + \hat{k}) = 10$	
	$\Rightarrow 4+3\lambda-2-4\lambda+7+2\lambda = 10$	
	$\Rightarrow \lambda + 9 = 10 \Rightarrow \lambda = 1$	1
	$\therefore$ Point of intersection is (7, 6, 9)	1
	Distance of (7, 6, 9) from (1, –2, 9) is	
	$d = \sqrt{(7-1)^2 + (6+2)^2 + 0}$	
	$= \sqrt{36+64} = 10$	1

		4
14.	**Case Study Problem :** A shopkeeper sells three types of flower seeds A1, A2, A3. They are sold in the form of a mixture, where the proportions of these seeds are 4 : 4 : 2, respectively. The germination rates of the three types of seeds are 45%, 60% and 35% respectively.  Based on the above information : (a) Calculate the probability that a randomly chosen seed will germinate; (b) Calculate the probability that the seed is of type A2, given that a randomly chosen seed germinates.	2 2
Sol.	E_1 : Seed is of type A_1 E_2 : Seed is of type A_2 E_3 : Seed is of type A_3 A: Seed germinates $P(E_1) = \frac{4}{10}, P(E_2) = \frac{4}{10}, P(E_3) = \frac{2}{10}$ $P(A\|E_1) = \frac{45}{100}, P(A\|E_2) = \frac{60}{100}, P(A\|E_3) = \frac{35}{100}$	½ ½ ½
	(a) $P(A) = P(E_1)P(A\|E_1) + P(E_2)P(A\|E_2) + P(E_3)P(A\|E_3)$ $= \frac{4}{10}\times\frac{45}{100} + \frac{4}{10}\times\frac{60}{100} + \frac{2}{10}\times\frac{35}{100}$ $= \frac{180+240+70}{1000}$ $= \frac{490}{1000} = \frac{49}{100}$ (b) $P\left(\frac{E_2}{A}\right) = \frac{P(E_2)P(A\|E_2)}{P(A)}$ $= \frac{\frac{4}{10}\times\frac{60}{100}}{\frac{490}{1000}} = \frac{24}{49}$	1 ½ ½+½ 4

MARKING SCHEME

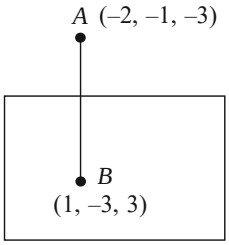
Senior Secondary School Examination TERM–II, 2022

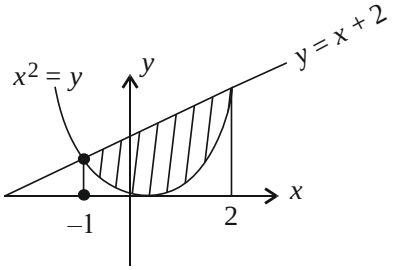
MATHEMATICS (Subject Code–041)

[Paper Code : 65/4/1]

Maximum Marks : 40

Q. No.	EXPECTED ANSWER / VALUE POINTS	Marks
	SECTION—A **Question Nos. 1 to 6 carry 2 marks each.**	
Q1.	A bag contains 3 red and 4 white balls. Three balls are drawn at random, one-by-one without replacement from the bag. If the first ball drawn is red in colour, then find the probability that the remaining two balls drawn are also red in colour.	**2**
A1.	A : First ball drawn is red B : Remaining two balls are red $P(B/A) = \dfrac{P(B \cap A)}{P(A)}$ $= \dfrac{\frac{1}{35}}{\frac{3}{7}} = \dfrac{1}{15}$	½ 1+½
Q2.	A coin is tossed twice. The following table shows the probability distribution of number of tails : X | 0 | 1 | 2 P(X) | K | 6K | 9K (a) Find the value of K. (b) Is the coin tossed biased or unbiased ? Justify your answer.	**2**
A2.	$16K = 1$ Let $P(T) = p$ and $P(H) = q$ **(a)** $K = \dfrac{1}{16}$ **(b)** $P(0) = \dfrac{1}{16}$, $P(1) = \dfrac{6}{16}$, $P(2) = \dfrac{9}{16}$ $q = \dfrac{1}{4} \Rightarrow p = \dfrac{3}{4}$ Since $P(H) \neq P(T)$, the coin, so tossed, is biased.	1 ½ ½

Q3.	The foot of a perpendicular drawn from the point (– 2, – 1, – 3) on a plane is (1, – 3, 3). Find the equation of the plane.	**2**
A3.	D.R.'s of normal vector are $\langle 3,-2,6\rangle$ Equation of plane is : $3(x-1)-2(y+3)+6(z-3)=0$ $\Rightarrow 3x-2y+6z=27$  	1 1
Q4.	(a) If $\lvert \vec{a} \times \vec{b} \rvert^2 + \lvert \vec{a} \cdot \vec{b} \rvert^2 = 400$ and $\lvert \vec{b} \rvert = 5$, then find the value of $\lvert \vec{a} \rvert$. **OR** (b) Find all the possible vectors of magnitude $5\sqrt{3}$ which are equally inclined to the coordinate axes.	**2**
A4.	**(a)** $\lvert \vec{a} \rvert^2 \lvert \vec{b} \rvert^2 \sin^2\theta + \lvert \vec{a} \rvert^2 \lvert \vec{b} \rvert^2 \cos^2\theta = 400$ $\lvert \vec{a} \rvert^2 \cdot 25(1) = 400$ $\lvert \vec{a} \rvert^2 = 16$ $\lvert \vec{a} \rvert = 4$ **Or** **(b)** Let the required vector be $x\hat{i}+x\hat{j}+x\hat{k}$ $\sqrt{3x^2} = 5\sqrt{3}$ $x^2 = 25 \Rightarrow x = \pm 5$ Required vectors are $5\hat{i}+5\hat{j}+5\hat{k}$ or $-5\hat{i}-5\hat{j}-5\hat{k}$.	1 ½ ½ ½ ½ ½ ½
Q5.	Find the general solution of the differential equation $\sec^2 x \cdot \tan y\, dx + \sec^2 y \cdot \tan x\, dy = 0.$	**2**
A5.	Given differential equation can be written as $\frac{\sec^2 x}{\tan x}dx + \frac{\sec^2 y}{\tan y}dy = 0$ Integrating, $\log\lvert\log x\rvert + \log\lvert\tan y\rvert = \log C$ $\tan x \cdot \tan y = C$	1 ½ ½

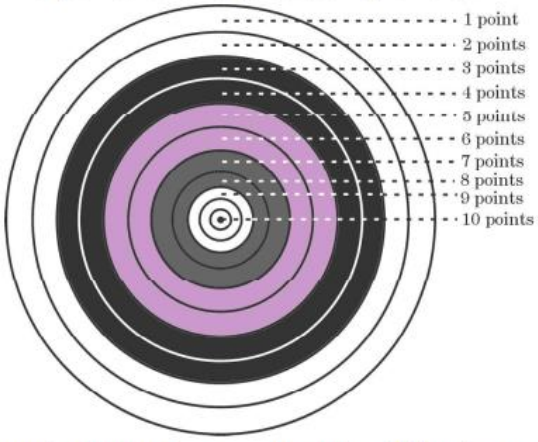
Q6.	Evaluate : $\int_0^1 x^2 e^x \, dx$	**2**
A6.	$\int_0^1 x^2 e^x dx = \left[x^2 e^x \right]_0^1 - \int_0^1 2xe^x dx$ $= \left[x^2 e^x - 2xe^x + 2e^x \right]_0^1$ $= e - 2$	1 ½ ½
	SECTION—B **Question Nos. 7 to 10 carry *3* marks each.**	
Q7.	Find the area of the region $\{(x, y) : x^2 \le y \le x + 2\}$, using integration.	**3**
A7.	 *(for correct figure)* *x*-coordinates of points of intersection are –1, 2. Required area = $\int_{-1}^{2} [(x+2) - x^2] dx$ $= \frac{(x+2)^2}{2} - \frac{x^3}{3}\Big|_{-1}^{2}$ $= \frac{16}{3} - \frac{5}{6} = \frac{9}{2}$	1 ½ ½ ½ ½
Q8.	(a) Find : $\int \frac{1}{e^x + 1} dx$ **OR** (b) Evaluate : $\int_1^4 \{ \lvert x \rvert + \lvert 3 - x \rvert \} dx$	**3**

A8.	**(a)** Let, $e^x = t \quad , \quad e^x dx = dt$	½
	$\int \frac{1}{e^x+1} dx = \int \frac{1}{t(t+1)} dt$	½
	$= \int \frac{dt}{t} - \int \frac{dt}{t+1}$	½
	$= \log \lvert t \rvert - \log \lvert t+1 \rvert + C$	1
	$= \log e^x - \log (e^x+1) + C$ or $\log \left\lvert \frac{e^x}{1+e^x} \right\rvert + C$	½
	Or	
	(b) $I = \int_1^4 \{\lvert x \rvert + \lvert 3-x \rvert\} dx$	
	$= \int_1^3 \{\lvert x \rvert + \lvert 3-x \rvert\} dx + \int_3^4 \{\lvert x \rvert + \lvert 3-x \rvert\} dx$	1
	$= \int_1^3 3 dx + \int_3^4 (2x-3) dx$	1
	$= \lvert 3x \rvert_1^3 + \lvert x^2 - 3x \rvert_3^4$	½
	= 6 + 4 = 10	½
Q9.	If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are mutually perpendicular vectors of equal magnitude, then prove that the vector $(2\vec{a} + \vec{b} + 2\vec{c})$ is equally inclined to both $\vec{a}$ and $\vec{c}$. Also, find the angle between $\vec{a}$ and $(2\vec{a} + \vec{b} + 2\vec{c})$.	**3**
A9.	Let, $\lvert \vec{a} \rvert = \lvert \vec{b} \rvert = \lvert \vec{c} \rvert = m$	½
	$\lvert 2\vec{a} + \vec{b} + 2\vec{c} \rvert^2 = 4 \lvert \vec{a} \rvert^2 + \lvert \vec{b} \rvert^2 + 4 \lvert \vec{c} \rvert^2 + 4\vec{a} \cdot \vec{b} + 4\vec{b} \cdot \vec{c} + 8\vec{a} \cdot \vec{c}$	
	$= 9m^2$	½
	$\lvert 2\vec{a} + \vec{b} + 2\vec{c} \rvert = 3m$	
	Let θ be angle between $2\vec{a} + \vec{b} + 2\vec{c}$ and $\vec{a}$.	
	$\cos\theta = \frac{(2\vec{a} + \vec{b} + 2\vec{c}) \cdot \vec{a}}{\lvert 2\vec{a} + \vec{b} + 2\vec{c} \rvert \lvert \vec{a} \rvert} = \frac{2m^2}{3m \cdot m} = \frac{2}{3}$	1
	Let, α be angle between $2\vec{a} + \vec{b} + 2\vec{c}$ and $\vec{c}$	

	Similarly, $\cos\alpha = \frac{2m^2}{3m \cdot m} = \frac{2}{3}$	
	$\therefore \cos\theta = \cos\alpha \Rightarrow \theta = \alpha$	½
	$\therefore$ Required angle = $\cos^{-1}\left(\frac{2}{3}\right)$	½
Q10.	(a) If a line makes 60° and 45° angles with the positive directions of x-axis and z-axis respectively, then find the angle that it makes with the positive direction of y-axis. Hence, write the direction cosines of the line. **OR** (b) Check whether the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-4}{5} = \frac{y-1}{2} = z$ are skew or not.	**3**
A10.	**(a)** $l = \cos 60° = \frac{1}{2}$, $n = \cos 45° = \frac{1}{\sqrt{2}}$	½+½
	Now, $l^2 + m^2 + n^2 = 1$	½
	$\Rightarrow \frac{1}{4} + m^2 + \frac{1}{2} = 1$	
	$\Rightarrow m^2 = \frac{1}{4} \Rightarrow m = \pm\frac{1}{2}$	½
	$\theta = 60°$	
	Required direction cosines are $\left\langle \frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}} \right\rangle$	1
	Or	
	(b) Let, $\vec{a}_1 = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{a}_2 = 4\hat{i} + \hat{j}$	
	$\vec{b}_1 = 2\hat{i} + 3\hat{j} + 4\hat{k}$, $\vec{b}_2 = 5\hat{i} + 2\hat{j} + \hat{k}$	
	$\vec{a}_2 - \vec{a}_1 = 3\hat{i} - \hat{j} - 3\hat{k}$	½
	$\vec{b}_1 \times \vec{b}_2 = -5\hat{i} + 18\hat{j} - 11\hat{k}$	1½
	Here, $(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = -15 - 18 + 33 = 0$	½
	Hence given lines are **not** skew lines.	½

	SECTION—C **Question Nos. 11 to 14 carry 4 marks each.**	
Q11.	Find the equations of the planes passing through the line of intersection of the planes $\vec{r}.(\hat{i}+3\hat{j})=6$ and $\vec{r}.(3\hat{i}-\hat{j}-4\hat{k})=0$, which are at a distance of 1 unit from the origin.	**4**
Q11.	Let the equations of required planes are: $\vec{r}\cdot\left(\hat{i}+3\hat{j}\right)-6+\lambda\left[\vec{r}\cdot\left(3\hat{i}-\hat{j}-4\hat{k}\right)\right]=0$ or $\vec{r}\cdot[(1+3\lambda)\hat{i}+(3-\lambda)\hat{j}-4\lambda\hat{k}]=6$ for some λ According to given condition $\frac{6}{\sqrt{(1+3\lambda)^2+(3-\lambda)^2+(-4\lambda)^2}}=1$ $36=1+9\lambda^2+6\lambda+9+\lambda^2-6\lambda+16\lambda^2$ $\Rightarrow 26\lambda^2=26 \Rightarrow \lambda=\pm1$ Equations of planes are: $\vec{r}\cdot(4\hat{i}+2\hat{j}-4\hat{k})=6$, $\vec{r}\cdot(-2\hat{i}+4\hat{j}+4\hat{k})=6$	1 1 1 ½+½
Q12.	(a) Find the particular solution of the differential equation $x\frac{dy}{dx}+y+\frac{1}{1+x^2}=0$, given that $y(1)=0$. **OR** (b) Find the general solution of the differential equation $x(y^3+x^3)\,dy=(2y^4+5x^3y)\,dx$.	**4**
A12.	**(a)** Given differential equation can be written as $\frac{dy}{dx}+\frac{1}{x}y=\frac{-1}{x(1+x^2)}$ I.F. $=e^{\int\frac{1}{x}dx}=e^{\log x}=x$ Solution is $y\cdot x=\int\frac{-1}{1+x^2}dx+C$ $\Rightarrow xy=-\tan^{-1}x+C$ Now $y(1)=0\Rightarrow C=\frac{\pi}{4}$ $\therefore$ Particular solution is $xy=\frac{\pi}{4}-\tan^{-1}x$	½ 1 1 1 ½
	Or **(b)** Given differential equation can be written as $\frac{dy}{dx}=\frac{2y^4+5x^3y}{xy^3+x^4}$ …(1)	½

	Let, $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$	½
	Equation (1) becomes $v + x\frac{dv}{dx} = \frac{2v^4 + 5v}{v^3 + 1}$	
	$\Rightarrow x\frac{dv}{dx} = \frac{v^4 + 4v}{v^3 + 1}$	
	$\frac{v^3 + 1}{v^4 + 4v}dv = \frac{dx}{x}$	1
	$\int \frac{4v^3 + 4}{v^4 + 4v}dv = 4\int \frac{dx}{x}$	½
	$\log\lvert v^4 + 4v\rvert = \log(x)^4 + \log C$	1
	$\log\left\lvert\frac{y^4 + 4yx^3}{x^4}\right\rvert = \log Cx^4$	
	$y^4 + 4yx^3 = Cx^8$	½
Q13.	Evaluate : $\int_0^{\pi/2} (2\log\cos x - \log\sin 2x)\,dx$	**4**
A13.	$\int_0^{\frac{\pi}{2}} (2\log\cos x - \log\sin 2x)\,dx$	
	$= \int_0^{\frac{\pi}{2}} \log\left(\frac{\cos^2 x}{2\sin x\cos x}\right)dx$	1
	$= \int_0^{\frac{\pi}{2}} \log\left(\frac{\cot x}{2}\right)dx = I$ (say) … (1)	½
	$I = \int_0^{\frac{\pi}{2}} \log\frac{\tan x}{2}\,dx$ (using property) … (2)	1
	Adding (1) and (2)	
	$2I = \int_0^{\frac{\pi}{2}} \log\left(\frac{\tan x}{2}\frac{\cot x}{2}\right)dx$	½
	$2I = \log\left(\frac{1}{4}\right)[x]_0^{\pi/2}$	½
	$I = \frac{\pi}{4}\log\frac{1}{4}$ or $-\frac{\pi}{2}\log 2$	½

Q14.	In a game of Archery, each ring of the Archery target is valued. The centremost ring is worth 10 points and rest of the rings are allotted points 9 to 1 in sequential order moving outwards. Archer A is likely to earn 10 points with a probability of 0·8 and Archer B is likely the earn 10 points with a probability of 0·9.  Based on the above information, answer the following questions : If both of them hit the Archery target, then find the probability that (a) exactly one of them earns 10 points. (b) both of them earn 10 points.	**4**
A14.	$P(A) = 0{\cdot}8$, $P(B) = 0{\cdot}9$ (a) P (exactly one of them earns 10 points) = $P(A)P(\bar{B}) + P(\bar{A})P(B)$ $= 0{\cdot}8 \times 0{\cdot}1 + 0{\cdot}2 \times 0{\cdot}9$ $= 0{\cdot}26$ (b) $P(A \cap B) = P(A)P(B)$ $= 0{\cdot}8 \times 0{\cdot}9$ $= 0{\cdot}72$	1½ ½ 1 1

MARKING SCHEME

Senior Secondary School Examination TERM–II, 2022

MATHEMATICS (Subject Code–041)

[Paper Code : 65/5/1]

Maximum Marks : 40

Q. No.	EXPECTED ANSWER / VALUE POINTS	Marks
	SECTION—A **Question Nos. 1 to 6 carry 2 marks each.**	
Q1.	Find : $\int \frac{dx}{\sqrt{4x-x^2}}$	**2**
A1.	$\int \frac{dx}{\sqrt{4x-x^2}}$ $= \int \frac{dx}{\sqrt{2^2-(x-2)^2}}$ $= \sin^{-1}\left(\frac{x-2}{2}\right)+C$	1 1
Q2.	Find the general solution of the following differential equation : $\frac{dy}{dx} = e^{x-y} + x^2e^{-y}$	**2**
A2.	$\int \frac{dy}{e^{-y}} = \int (x^2+e^x)\,dx$ $e^y = \frac{x^3}{3}+e^x+C$	1 1
Q3.	Let X be a random variable which assumes values x_1, x_2, x_3, x_4 such that $2P(X=x_1) = 3P(X=x_2) = P(X=x_3) = 5P(X=x_4)$. Find the probability distribution of X.	**2**
A3.	Let $P(X=x_3)=k$ $P(X=x_1)=\frac{k}{2},\ P(X=x_2)=\frac{k}{3},\ P(X=x_4)=\frac{k}{5}$ $\frac{k}{2}+\frac{k}{3}+k+\frac{k}{5}=1$ $k=\frac{30}{61}$	½ ½

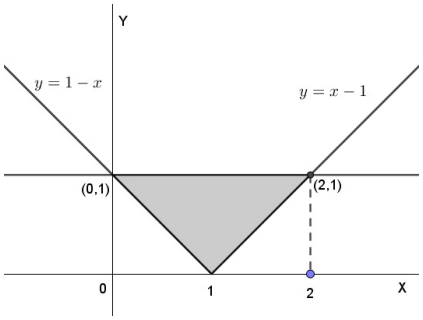
	Probability Distribution of X is: X | x_1 | x_2 | x_3 | x_4 $P(X)$ | 15/61 | 10/61 | 30/61 | 6/61	1
Q4.	If $\vec{a} = \hat{i}+\hat{j}+\hat{k}$, $\vec{a}.\vec{b}=1$ and $\vec{a}\times\vec{b} = \hat{j}-\hat{k}$, then find $\left\|\vec{b}\right\|$	**2**
A4.	Let $\vec{b} = x\hat{i}+y\hat{i}+z\hat{k}$ $\vec{a}\cdot\vec{b}=1 \Rightarrow x+y+z=1$ ... (1) ½ $\vec{a}\times\vec{b} = \hat{j}-\hat{k}$ $\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & 1 & 1\\ x & y & z\end{vmatrix} = \hat{i}(z-y)-\hat{j}(z-x)+\hat{k}(y-x) = \hat{j}-\hat{k}$ ½ $z-y=0 \Rightarrow y=z$ $\Rightarrow x-z=1$ ... (2) $x-y=1$ ... (3) Solving (1), (2), (3) $x=1,\ y=0,\ z=0$ ½ $\vec{b}=\hat{i}$, so $\|\vec{b}\|=1$ ½	
Q5.	If a line makes an angle α, β, γ with the coordinate axes, then find the value of $\cos2\alpha+\cos2\beta+\cos2\gamma$.	**2**
A5.	$l=\cos\alpha,\ m=\cos\beta,\ n=\cos\gamma \Rightarrow \cos^2\alpha+\cos^2\beta+\cos^2\gamma=1$ 1 $\cos2\alpha+\cos2\beta+\cos2\gamma$ $=2\cos^2\alpha-1+2\cos^2\beta-1+2\cos^2\gamma-1$ $=2(\cos^2\alpha+\cos^2\beta+\cos^2\gamma)-3$ $=2-3$ $=-1$ 1	

Q6.	(a) Events A and B are such that $P(A)=\frac{1}{2}$, $P(B)=\frac{7}{12}$ and $P(\overline{A}\cup\overline{B})=\frac{1}{4}$ Find whether the events A and B are independent or not. **OR** (b) A box B_1 contains 1 white ball and 3 red balls. Another box B_2 contains 2 white balls and 3 red balls. If one ball is drawn at random from each of the boxes B_1 and B_2, then find the probability that the two balls drawn are of the same colour.	**2**
A6(a).	$P(A)=\frac{1}{2},\ P(B)=\frac{7}{12},\ P(\overline{A}\cup\overline{B})=\frac{1}{4}$ $P(\overline{A}\cup\overline{B})=P(\overline{A\cap B})=1-P(A\cap B)$ $P(A\cap B)=1-\frac{1}{4}=\frac{3}{4}$ $P(A)\times P(B)=\frac{1}{2}\times\frac{7}{12}=\frac{7}{24}$ $P(A)\times P(B)\neq P(A\cap B)$ $\therefore$ A and B are not independent	½ ½ ½ ½
A6.(b)	**Or** P (both balls drawn are of same colour) $= P$ (both white) + P (both red) $=\frac{1}{4}\times\frac{2}{5}+\frac{3}{4}\times\frac{3}{5}=\frac{11}{20}$	1 1
	SECTION—B **Question Nos. 7 to 10 carry *3* marks each.**	
Q7.	Evaluate : $\int_0^{\pi/4}\frac{dx}{1+\tan x}$	**3**
A7.	$I=\int_0^{\pi/4}\frac{dx}{1+\tan x}$ $I=\int_0^{\pi/4}\frac{dx}{1+\tan\left(\frac{\pi}{4}-x\right)}$ (using property) $I=\int_0^{\pi/4}\frac{dx}{1+\frac{1-\tan x}{1+\tan x}}$	½ ½

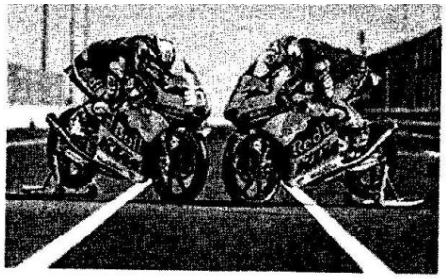
	$I = \frac{1}{2}\int_0^{\pi/4}(1+\tan x)\,dx$	½
	$I = \frac{1}{2}\left[x+\log\sec x\right]_0^{\pi/4}$	1
	$= \frac{1}{2}\left[\frac{\pi}{4}+\log\sqrt{2}\right] = \frac{1}{2}\left(\frac{\pi}{4}+\frac{1}{2}\log 2\right)$	
	$= \frac{\pi}{8}+\frac{1}{4}\log 2$	½
Q8.	(a) If $\vec{a}$ and $\vec{b}$ are two vectors such that $\left\|\vec{a}+\vec{b}\right\| = \left\|\vec{b}\right\|$, then prove that $\left(\vec{a}+2\vec{b}\right)$ is perpendicular to $\vec{a}$. **OR** (b) If $\vec{a}$ and $\vec{b}$ are unit vectors and θ is the angle between them, then prove that $\sin\frac{\theta}{2} = \frac{1}{2}\left\|\vec{a}-\vec{b}\right\|$.	**3**
A8.(a)	$\|\vec{a}+\vec{b}\| = \|\vec{b}\|$	
	$\left(\vec{a}+\vec{b}\right)^2 = \left(\vec{b}\right)^2$	1
	$\vec{a}^2+\vec{b}^2+2\vec{a}.\vec{b} = \vec{b}^2$	½
	$\vec{a}^2+2\vec{a}.\vec{b} = 0$	
	$(\vec{a}+2\vec{b}).\vec{a} = 0$	1
	$\therefore (\vec{a}+2\vec{b}) \perp \vec{a}$	½
	Or	
A8.(b)	Consider	
	$\|\vec{a}-\vec{b}\|^2 = (\vec{a}-\vec{b}).(\vec{a}-\vec{b}) = \|\vec{a}\|^2 - 2\vec{a}.\vec{b} + \|\vec{b}\|^2$	1
	$= 1-2\|\vec{a}\|\|\vec{b}\|\cos\theta+1$	
	$= 2-2\cos\theta$	1
	$= 2\left(2\sin^2\frac{\theta}{2}\right)$	
	$\therefore \sin\frac{\theta}{2} = \frac{1}{2}\|\vec{a}-\vec{b}\|$	1

Q9.	Find the equation of the plane passing through the line of intersection of the planes $\vec{r}\cdot\left(\hat{i}+\hat{j}+\hat{k}\right)=10$ and $\vec{r}\cdot\left(2\hat{i}+3\hat{j}-\hat{k}\right)+4=0$ and passing through the point (–2, 3, 1).	**3**
A9.	Equation of plane through the intersection of given two planes is $\vec{r}\cdot[(\hat{i}+\hat{j}+\hat{k})+\lambda(2\hat{i}+3\hat{j}-\hat{k})]-10+4\lambda=0$ $\vec{r}\cdot[(1+2\lambda)\hat{i}+(1+3\lambda)\hat{j}+(1-\lambda)\hat{k}]-10+4\lambda=0$ Point (–2, 3, 1) lies on it. $\therefore -2-\cancel{4\lambda}+3+9\lambda+1-\lambda-10+\cancel{4\lambda}=0$ $8\lambda=8\Rightarrow\lambda=1$ $\therefore$ Equation of plane is $\vec{r}.(3\hat{i}+4\hat{j})-6=0$	1 1 1
Q10.	(a) Find : $\int e^x.\sin 2x\,dx$ **OR** (b) Find : $\int\frac{2x}{(x^2+1)(x^2+2)}dx$	**3**
A10.(a)	$I=\int_{II}e^x\sin 2x\,dx$ $I=\sin 2x\,e^x-\int 2\cos 2x\,e^x\,dx$ $=e^x\sin 2x-2[\cos 2x\,e^x-\int(-2\sin 2x)e^x dx]$ $I=e^x\sin 2x-2\cos 2x\,e^x-4I$ $5I=e^x\sin 2x-2\cos 2xe^x$ $\therefore I=\frac{1}{5}e^x[(\sin 2x-2\cos 2x)]+C$	1 ½ ½ ½ ½
	Or	
A10.(b)	$\int\frac{2x}{(x^2+1)(x^2+2)}dx$ Let $x^2=t,\ 2x\,dx=dt$ $\therefore\int\frac{2x}{(x^2+1)(x^2+2)}dx=\int\frac{dt}{(t+1)(t+2)}$	½

	Getting $\frac{1}{(t+1)(t+2)}=\frac{1}{t+1}+\frac{-1}{t+2}$ (by Partial Fraction) $\int\frac{dt}{(t+1)(t+2)}=\int\frac{dt}{t+1}-\int\frac{dt}{t+2}$ $=\log\lvert t+1\rvert-\log\lvert t+2\rvert+C$ $=\log\left(x^2+1\right)-\log\left(x^2+2\right)+C$ or $\log\left(\frac{x^2+1}{x^2+2}\right)+C$	1 1 ½
	SECTION—C **Question Nos. 11 to 14 carry 4 marks each.**	
Q11.	Three persons A, B and C apply for a job of manager in a private company. Chances of their selection are in the ratio 1 : 2 : 4. The probability that A, B and C can introduce changes to increase the profits of a company are 0.8, 0.5 and 0.3 respectively. If increase in the profit does not take place, find the probability that it is due to the appointment of A.	**4**
A11.	E_1 : A is selected E_2 : B is selected E_3 : C is selected F : increase in profit does not take place $P(E_1)=1/7,\ P(E_2)=2/7,\ P(E_3)=4/7$ $P(F\mid E_1)=0{\cdot}2,\ P(F\mid E_2)=0{\cdot}5,\ P(F\mid E_3)=0{\cdot}7$ $P(E_1\mid F)=\frac{P(E_1)P(F\mid E_1)}{P(E_1)P(F\mid E_1)+P(E_2)P(F\mid E_2)+P(E_3)P(F\mid E_3)}$ $=\frac{\frac{1}{7}\times\frac{2}{10}}{\frac{1}{7}\times\frac{2}{10}+\frac{2}{7}\times\frac{5}{10}+\frac{4}{7}\times\frac{7}{10}}$ $=\frac{2}{40}=\frac{1}{20}$	½ 1 1 1 ½
Q12.	Find the area bounded by the curves $y=\lvert x-1\rvert$ and $y=1$, using integration.	**4**

A12.	 *(for correct figure)*	1
	Area of the bounded region is	
	$\int_0^1 [1-(1-x)]\,dx + \int_1^2 [1-(x-1)]\,dx$	1
	$\int_0^1 x\,dx + \int_1^2 (2-x)\,dx$	
	$= \frac{x^2}{2}\Big]_0^1 + \left[2x - \frac{x^2}{2}\right]_1^2$	1
	$= \frac{1}{2} + \left[4 - \frac{4}{2} - 2 + \frac{1}{2}\right]$	
	$= -1 + 2 = 1$	1
Q13.	(a) Solve the following differential equation : $(y - \sin^2 x)dx + \tan x\,dy = 0$	**4**
	OR	
	(b) Find the general solution of the differential equation : $(x^3 + y^3)dy = x^2 y\,dx$	
A13.(a)	$(y - \sin^2 x)\,dx + \tan x\,dy = 0$	
	$\frac{dy}{dx} + \frac{y}{\tan x} = \frac{\sin^2 x}{\tan x}$	1
	$\frac{dy}{dx} + (\cot x)\,y = \sin x \cos x$	
	$I.F. = e^{\int \cot x\,dx} = e^{\log(\sin x)} = \sin x$	1
	Solution is given by	
	$y \sin x = \int \sin^2 x \cos x\,dx$	½

A13.(b)	$y\sin x = \int t^2 dt \qquad \because \sin x = t\ ,\ \cos x\, dx = dt$	½
	$y\sin x = \frac{t^3}{3} + C$	½
	$y\sin x = \frac{\sin^3 x}{3} + C$	½
	Or	
	$\frac{dy}{dx} = \frac{x^2 y}{x^3 + y^3}$	½
	Put $y = vx,\ \frac{dy}{dx} = v + x\frac{dv}{dx}$	½
	$v + x\frac{dv}{dx} = \frac{v}{1+v^3} \Rightarrow x\frac{dv}{dx} = \frac{-v^4}{1+v^3}$	
	$\int \frac{1+v^3}{v^4} dv = -\int \frac{dx}{x}$	1
	$\int \frac{1}{v^4} dv + \int \frac{1}{v} dv = -\log\lvert x\rvert + C$	
	$\frac{1}{-3v^3} + \log\lvert v\rvert = -\log\lvert x\rvert + C$	1½
	$\frac{-x^3}{3y^3} + \log\left\lvert\frac{y}{x}\right\rvert = -\log\lvert x\rvert + C$ or $\frac{-x^3}{3y^3} + \log\lvert y\rvert = C$	½

Q14.	Two motorcycles A and B are running at the speed more than the allowed speed on the roads represented by the lines $\vec{r} = \lambda\left(\hat{i}+2\hat{j}-\hat{k}\right)$ and $\vec{r} = \left(3\hat{i}+3\hat{j}\right)+\mu\left(2\hat{i}+\hat{j}+\hat{k}\right)$ respectively.  Based on the above information, answer the following questions : (a) Find the shortest distance between the given lines. (b) Find the point at which the motorcycles may collide.	**4**
A14.(a)	$\vec{a}_1 = 0\hat{i}+0\hat{j}+0\hat{k}$, $\vec{a}_2 = 3\hat{i}+3\hat{j}$ $\vec{a}_2-\vec{a}_1 = 3\hat{i}+3\hat{j}$ $\vec{b}_1\times\vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -1 \\ 2 & 1 & 1 \end{vmatrix}$ $= 3\hat{i}-3\hat{j}-3\hat{k}$ $\text{SD} = \dfrac{\left\lvert(\vec{a}_2-\vec{a}_1)\cdot(\vec{b}_1\times\vec{b}_2)\right\rvert}{\lvert\vec{b}_1\times\vec{b}_2\rvert}$ Now, $(\vec{a}_2-\vec{a}_1)\cdot(\vec{b}_1\times\vec{b}_2) = (3\hat{i}+3\hat{j})(3\hat{i}-3\hat{j}-3\hat{k})$ $= 9-9 = 0$ Shortest distance between two lines = 0	½ 1 ½
A14.(b)	Any point on the line $\vec{r} = \lambda(\hat{i}+2\hat{j}-\hat{k})$ is $\lambda\hat{i}+2\lambda\hat{j}-\lambda\hat{k}$ Any point on the line $\vec{r} = 3\hat{i}+3\hat{j}+\mu(2\hat{i}+\hat{j}+\hat{k})$ is $(2\mu+3)\hat{i}+(\mu+3)\hat{j}+\mu\hat{k}$ As the lines are intersecting, $\lambda = 2\mu+3, 2\lambda = \mu+3$ On solving $\mu = -1,\ \lambda = 1$ Point of intersection is $\hat{i}+2\hat{j}-\hat{k}$ or (1, 2, –1)	½ ½ ½ ½

MARKING SCHEME

Senior Secondary School Examination TERM–II, 2022

MATHEMATICS (Subject Code–041B)

[Paper Code: 65B]

Maximum Marks: 40

***Instructions*:**

1. Instructions for drawing up the MS should be followed carefully.
2. If general instructions have to be given, do so, at the beginning of the page itself.
3. Some subjects will require specific directions for particular type of questions. Give these at the beginning of the concerned question.
4. Do indicate the value points, time required for the questions.
5. Write page number of the topic from where question is asked.

Q. No.	EXPECTED ANSWER / VALUE POINTS	Marks
1(a)	Find : $\int \frac{1}{\sqrt{12+4x-x^2}}\,dx$	2
Sol.	$I = \int \frac{1}{\sqrt{12+4x-x^2}}\,dx$ $= \int \frac{1}{\sqrt{-(x^2-4x+4-16)}}\,dx = \int \frac{1}{\sqrt{16-(x-2)^2}}\,dx$ $= \sin^{-1}\frac{x-2}{4} + C$ Or	1 1 2
1(b)	Find : $\int \frac{xe^x}{(x+4)^5}\,dx$	2
Sol.	$I = \int \frac{xe^x}{(x+4)^5}\,dx = \int \frac{(x+4)-4}{(x+4)^5}e^x dx$ $= \int \left[\frac{1}{(x+4)^4} - \frac{4}{(x+4)^5}\right]e^x dx$ $= \int [f(x)+f'(x)]\,e^x dx = f(x)\cdot e^x + C \quad \left(\text{where } f(x)=\frac{1}{(x+4)^4} \Rightarrow f'(x)=\frac{-4}{(x+4)^5}\right)$ $= \frac{1}{(x+4)^4}e^x + C$	½ ½ 1 2
2.	Find the general solution of the following differential equation : $(4 + y^2)(3 + \log x)\,dx + x\,dy = 0$	2

Sol.	Given diff. equation can be written as $\frac{(3+\log x)}{x}dx+\frac{1}{4+y^2}dy=0$ Integrating to get $\frac{1}{2}(3+\log x)^2+\frac{1}{2}\tan^{-1}\frac{y}{2}=C_1$ Or, $(3+\log x)^2+\tan^{-1}\frac{y}{2}=C$	1 1 2
3.	If $\lvert\vec{a}\rvert = 3$, $\lvert\vec{b}\rvert = 2\sqrt{3}$ and $\vec{a}\cdot\vec{b} = 6$, then find the value of $\lvert\vec{a}\times\vec{b}\rvert$.	2
Sol.	$6=\lvert\vec{a}\rvert\lvert\vec{b}\rvert\cos\theta \Rightarrow 6=3\cdot 2\sqrt{3}\cdot\cos\theta$ $\Rightarrow \cos\theta=\frac{1}{\sqrt{3}}$ $\lvert\vec{a}\times\vec{b}\rvert=\lvert\vec{a}\rvert\lvert\vec{b}\rvert\sin\theta=(3)(2\sqrt{3})\sqrt{1-\frac{1}{3}}$ $=6\sqrt{3}\cdot\frac{\sqrt{2}}{\sqrt{3}}=6\sqrt{2}$	1 1 2
4.	Find the direction cosines of a line whose cartesian equation is given as $3x + 1 = 6y - 2 = 1 - z$.	2
Sol.	Given equation is $3x+1=6y-2=1-z$ $\Rightarrow \frac{x+\frac{1}{3}}{\frac{1}{3}}=\frac{y-\frac{1}{3}}{\frac{1}{6}}=\frac{z-1}{-1}$ $\Rightarrow$ DRs are $\left\langle\frac{1}{3},\frac{1}{6},-1\right\rangle$ or $<2, 1, -6>$ $\Rightarrow$ DCs are $\frac{2}{\sqrt{41}},\frac{1}{\sqrt{41}},-\frac{6}{\sqrt{41}}$ Or $-\frac{2}{\sqrt{41}},-\frac{1}{\sqrt{41}},\frac{6}{\sqrt{41}}$	1 ½ ½ 2
5	In a toss of three different coins, find the probability of coming up of three heads, if it is known that at least one head comes up.	2

Sol.

Let E_1 : getting three heads

E_2 : getting at least one head

$P(E_2) = 7/8$; $P(E_1 \cap E_2) = 1/8$ — ½ + ½

$$P(E_1 \mid E_2) = \frac{P(E_1 \cap E_2)}{P(E_2)} = \frac{\frac{1}{8}}{\frac{7}{8}}$$ — ½

$$= \frac{1}{7}$$ — ½

2

6. Two rotten apples are mixed with 8 fresh apples. Find the probability distribution of number of rotten apples, if two apples are drawn at random, one-by-one without replacement. — 2

Sol.

Rotten apples = 2, fresh = 8, total = 10

Let X be the random variable (no. of rotten apples)

X can take values 0, 1, 2 — ½

Probability Distribution is — 1½

X	0	1	2
$P(X)$	$\frac{{}^8C_2}{{}^{10}C_2} = \frac{28}{45}$	$\frac{{}^8C_1 \cdot {}^2C_1}{{}^{10}C_2} = \frac{16}{45}$	$\frac{{}^2C_2}{{}^{10}C_2} = \frac{1}{45}$

2

7. Evaluate : $\int_0^{\pi/3} |\cos 3x| \, dx$ — 3

Sol.

$$I = \int_0^{\pi/3} |\cos 3x| \, dx$$

$$= \int_0^{\pi/6} \cos 3x \, dx - \int_{\pi/6}^{\pi/3} \cos 3x \, dx$$ — 1½

$$= \frac{\sin 3x}{3}\Big]_0^{\pi/6} - \frac{\sin 3x}{3}\Big]_{\pi/6}^{\pi/3}$$ — ½

$$= \frac{1}{3}[1-0] - \frac{1}{3}[0-1] = \frac{2}{3}$$ — 1

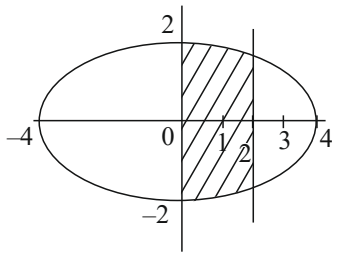
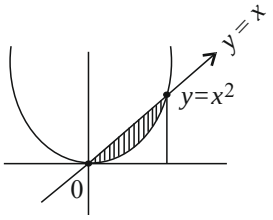
3

8.(a) Find the general solution of the following differential equation :

$$2x\, e^{y/x}\, dy + (x - 2y\, e^{y/x})\, dx = 0$$ — 3

Sol.	Given diff. equation is $\frac{dy}{dx} = \frac{2ye^{y/x} - x}{2xe^{y/x}}$ Put, $\frac{y}{x} = v \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$ $\therefore \quad v + x\frac{dv}{dx} = \frac{2ve^{v} - 1}{2e^{v}}$ $\Rightarrow x\frac{dv}{dx} = \frac{2ve^{v} - 1}{2e^{v}} - v = -\frac{1}{2e^{v}}$ $\Rightarrow 2e^{v}dv = -\frac{dx}{x}$ Integrating to get, $2e^{v} = -\log\|x\| + C$ $\Rightarrow 2e^{y/x} + \log\|x\| = C$	½ ½ ½ ½ 1
	Or	3
8.(b)	Find the particular solution of the differential equation $(2x^2 + y) \cdot \frac{dx}{dy} = x$; given that y = 2 when x = 1.	3
Sol.	Given diff. equation can be written as $x\frac{dy}{dx} - y = 2x^2 \ \ or \ \frac{dy}{dx} - \frac{1}{x}y = 2x$ IF $= e^{\int -\frac{1}{x}dx} = e^{-\log x} = e^{\log\frac{1}{x}} = \frac{1}{x}$ $\therefore$ Solution is $y \cdot \frac{1}{x} = \int 2x \cdot \frac{1}{x}dx = 2x + C$ $\Rightarrow y = 2x^2 + Cx$ when $x = 1,\ y = 2 \Rightarrow 2 = 2 + C \Rightarrow C = 0$ $\Rightarrow$ Particular Solution is $y = 2x^2$	1 ½ ½ ½ ½
		3
9.	Using vectors, find the area of the triangle with vertices A(–1, 0, – 2), B(0, 2, 1) and C(–1, 4, 1).	3
Sol.	Taking any two sides (say AB and AC) $\overrightarrow{AB} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\overrightarrow{AC} = 0\hat{i} + 4\hat{j} + 3\hat{k}$ $\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ 0 & 4 & 3 \end{vmatrix} = -6\hat{i} - 3\hat{j} + 4\hat{k}$	1 1

	Area of $\Delta = \frac{1}{2}\lvert \overrightarrow{AB} \times \overrightarrow{AC} \rvert = \frac{1}{2}\sqrt{36+9+16} = \frac{1}{2}\sqrt{61}$	1 3
10.(a)	Find the shortest distance between the lines $\vec{r} = (\lambda + 1)\,\hat{i} + (\lambda + 4)\,\hat{j} - (\lambda - 3)\,\hat{k}$, and $\vec{r} = (3 - \mu)\,\hat{i} + (2\mu + 2)\,\hat{j} + (\mu + 6)\,\hat{k}$.	3
Sol.	Lines are $\vec{r} = \hat{i} + 4\hat{j} + 3\hat{k} + \lambda(\hat{i} + \hat{j} - \hat{k})$ and $\vec{r} = 3\hat{i} + 2\hat{j} + 6\hat{k} + \mu(-\hat{i} + 2\hat{j} + \hat{k})$ $\vec{a}_2 - \vec{a}_1 = 2\hat{i} - 2\hat{j} + 3\hat{k}$ $\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -1 \\ -1 & 2 & 1 \end{vmatrix} = 3\hat{i} + 0\hat{j} + 3\hat{k}$ $\text{SD} = \frac{\lvert(\vec{a}_2 - \vec{a}_1)\cdot(\vec{b}_1 \times \vec{b}_2)\rvert}{\lvert \vec{b}_1 \times \vec{b}_2 \rvert} = \frac{6+9}{\sqrt{18}} = \frac{15}{3\sqrt{2}} = \frac{5}{\sqrt{2}}$ Or	½ ½ 1 1 3
10.(b)	Find the distance of the point P(4, 3, 2) from the plane determined by the points A(–1, 6, – 5), B(– 5, – 2, 3) and C(2, 4, – 5).	3
Sol.	Equation of plane is $\begin{vmatrix} x+1 & y-6 & z+5 \\ -4 & -8 & 8 \\ 3 & -2 & 0 \end{vmatrix} = 0$ $\Rightarrow 16(x+1) + 24(y-6) + 32(z+5) = 0$ $\Rightarrow 2x + 3y + 4z + 4 = 0$ Required Distance $= \frac{\lvert 8+9+8+4 \rvert}{\sqrt{4+9+16}} = \sqrt{29}$	1 1 1 3
11.	Find : $\int \frac{x^2 + x + 1}{(x+1)(x^2+4)}\,dx$	4
Sol.	$I = \int \frac{x^2+x+1}{(x+1)(x^2+4)}\,dx$ $\frac{x^2+x+1}{(x+1)(x^2+4)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+4} = \frac{\frac{1}{5}}{x+1} + \frac{\frac{4}{5}x + \frac{1}{5}}{x^2+4}$	½ + 1½

	$\Rightarrow I = \int \frac{\frac{1}{5}}{x+1}dx + \frac{2}{5}\int \frac{2x}{x^2+4}dx + \frac{1}{5}\int \frac{dx}{x^2+4}$ $= \frac{1}{5}\log\lvert x+1\rvert + \frac{2}{5}\log\lvert x^2+4\rvert + \frac{1}{10}\tan^{-1}\frac{x}{2} + C$	½ 1½ 4
12.(a)	Find the area bounded by the ellipse $x^2 + 4y^2 = 16$ and the ordinates $x = 0$ and $x = 2$, using integration.	4
Sol.	$A = 2\int_0^2 y\,dx$ $= 2\cdot\frac{1}{2}\cdot\int_0^2 \sqrt{16-x^2}\,dx$ $= \left[\frac{x}{2}\sqrt{16-x^2} + 8\sin^{-1}\frac{x}{4}\right]_0^2$ $= 2\sqrt{3} + 8\cdot\frac{\pi}{6} = \left(2\sqrt{3} + \frac{4\pi}{3}\right)$  **Or**	1 2 1 4
12.(b)	Find the area of the region $\{(x, y) : x^2 \le y \le x\}$, using integration.	4
Sol.	Getting x = 0, x = 1 for the point of intersection $A = \int_0^1 x\,dx - \int_0^1 x^2 dx$ $= \frac{x^2}{2} - \frac{x^3}{3}\Big]_0^1$ $= \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$  	1 1 1 1 4
13.	Find the coordinates of the point where the line through $(4, -3, -4)$ and $(3, -2, 2)$ crosses the plane $2x + y + z = 6$.	4
Sol.	Equation of line through (4, –3, –4) and (3, –2, 2) is $\frac{x-4}{-1} = \frac{y+3}{1} = \frac{z+4}{6}$ Any point on the line is $(-\lambda+4, \lambda-3, 6\lambda-4)$ If the point lies on the plane, then $2(-\lambda+4) + (\lambda-3) + 6\lambda - 4 = 6$ $\Rightarrow 5\lambda = 5 \Rightarrow \lambda = 1$	1 1 1

	∴ Point is $(3, -2, 2)$	1
		4
14.	A laboratory blood test is 98% effective in detecting a certain disease when it is in fact, present. However, the test also yields a false positive result for 0·4% of the healthy person tested. From a large population, it is given that 0·2% of the population actually has the disease. Based on the above, answer the following questions : (a) One person, from the population, is taken at random and given the test. Find the probability of his getting a positive test result. (b) What is the probability that the person actually has the disease, given that his test result is positive ?	4
Sol.	Let E_1 : *person has the disease* E_2 : *person does not have the disease* A : *getting a +ve test report*	1
	$P(E_1) = \frac{2}{1000}, \quad P(E_2) = \frac{998}{1000}$	½
	$P(A \mid E_1) = \frac{98}{100}, \quad P(A \mid E_2) = \frac{4}{1000}$	½
(a)	$P(A) = P(E_1)P(A \mid E_1) + P(E_2)P(A \mid E_2)$ $= \frac{2}{1000} \times \frac{98}{100} + \frac{998}{1000} \times \frac{4}{1000} = \frac{5952}{1000000}$	½ + ½
(b)	$P(E_1 \mid A) = \frac{P(E_1)P(A \mid E_1)}{\Sigma P(E_i)P(A \mid E_i)}$ $= \frac{\frac{2}{1000} \times \frac{980}{1000}}{\frac{5952}{1000000}} = \frac{1960}{5952} = \frac{245}{744}$	½ + ½
		4